

BioSpecInfo — Technical Dossier

Open-access web platform for computational chemistry and the interactive consultation of chemical data on any device.

Author	Samuele Pio Provenzano
Thesis advisor	Prof. Savino Longo — University of Bari Aldo Moro
Profile	Cheminformatics Engineer / Scientific Software Developer
Software type	Progressive Web App (PWA), client-side, local-first, offline-first
Live demo	https://samupropio1-ship-it.github.io/BioSpecInfo-v11/
Repository	github.com/samupropio1-ship-it/BioSpecInfo-v11

This dossier summarises the application and is the cover of the full technical documentation (SAD, Validation & Benchmark, Security & Compliance, Licensing) attached below.

1. Executive summary

BioSpecInfo is an **interactive cheminformatics platform** that runs all of its scientific logic **in the client browser**, with no server: molecular analysis, spectral prediction, 2D/3D modelling and educational tools — all **free, installable as an app and working offline**.

Its distinctive value is the **rare combination of advanced chemistry and modern software engineering** (WebAssembly, WebGL, PWA), with spectral predictors **quantitatively validated** against the official SDBS and NIST databases.

2. Product factsheet

Value proposition — A **≈4.5 MB** web platform (main file) that runs molecular analysis, spectral prediction and 3D visualisation at **zero server cost** and **100% offline-ready**.

Technical architecture	Content & capabilities
Client-side / Local-First (no backend)	RDKit cheminformatics engine (WebAssembly)
Single-file HTML5 / ES6, installable as PWA	Spectra Centre : IR, $^1\text{H}/^{13}\text{C}$ -NMR, MS + isotope pattern, UV-Vis, Raman
2D (SVG/Canvas) and 3D (three.js, 3Dmol.js) rendering	Organic Chemistry : 25 interactive modules
Service Worker offline-first, versioned	Retrosynthesis : 46 strategies + animated 2D/3D mechanisms
Zero copyleft dependencies (MIT/BSD-3 only)	3D Drug Atlas : 63 diseases (23 tumours), 3D organs, PubMed

Technical architecture	Content & capabilities
Local storage (localStorage, SQLite via sql.js)	3D astrochemistry, Quantum Lab, Academy (SM-2 quizzes)

Quantitative validation (thesis benchmark)

- ¹H-NMR vs SDBS → **MAE 0.31 ppm** (89% of signals within ±0.5 ppm)
- IR (C=O band) vs NIST WebBook → **MAE 11 cm⁻¹** (90% within ±30 cm⁻¹)
- Interactive performance **30–60 fps** from iPhone/iPad to Raspberry Pi 4

3. Verified capabilities (current code state)

Spectra Centre (molecular editor & analysis) — draw from scratch (full periodic table, templates, drag-to-bond) or load via SMILES/library →

- All chemical notations: formula, canonical SMILES, CXSMILES, SMARTS, InChI, InChIKey, mol block, common/IUPAC name, CAS (one-click copy).

- Predicted spectra with bibliographic sources; **isotope pattern** (IUPAC abundances) and MS **adduct masses**.

- Functional groups and properties (H donors/acceptors, aromatic, rotatable, stereocentres) **highlightable on the 2D and 3D structure**.

- Drug-likeness: **Lipinski, Veber, Egan, Ghose, Muegge**, ESOL solubility,

QED, Brenk/PAINS structural alerts, physicochemical radar.

- **Tanimoto** similarity (Morgan ECFP4 fingerprint), elemental analysis, SMARTS substructure search, SVG/PNG/MOL export.

Synthesis & Retrosynthesis — 46 strategies with an **animated mechanism player** in **2D and 3D**: *electron-pushing* arrows, reaction intermediates, and a retrosynthesis mode showing **synthon formation**.

Organic Chemistry — 25 modules (fundamentals → heterocycles, pericyclic & Woodward–Hoffmann, organometallics & cross-coupling, radicals, spectroscopy, retrosynthesis & protecting groups).

Data Science — 3D Drug Atlas (13 body regions, 63 diseases of which 23 tumours, 3D organ models, PubMed links), ML labs, SQL playground.

Other — 3D astrochemistry, Quantum Lab, Academy with spaced repetition (SM-2), interactive simulations, databases of elements/drugs/biomolecules.

Overall metrics declared in the thesis (historical data assets): 324 organic reactions in vector SVG graphics, 26 biosynthetic pathways, 140+ drugs, 80+ biomolecules, 118 elements, 10 quantum simulators.

4. Technology stack

HTML5 · ES6/ESNext · WebAssembly (RDKit MinimalLib, sql.js/SQLite) · WebGL (three.js, 3Dmol.js) · SVG / Canvas 2D · Service Worker / PWA · public API integration (PubChem PUG-REST, NIST, SDBS, MassBank, nmrshiftdb2, NASA/ESA).

5. Author's declaration (self-attestation)

The undersigned **Samuele Pio Provenzano** declares that:

1. The software **BioSpecInfo** is the author's **original work**, except for the third-party open-source libraries (RDKit, 3Dmol.js, three.js, SmilesDrawer, sql.js) used under their respective permissive licenses and documented in the *Open-Source Licensing & IP Matrix*.

1. The reported validation data (MAE 0.31 ppm for ^1H -NMR vs SDBS; MAE 11 cm^{-1} for the IR C=O band vs NIST) were measured by the author as part of the thesis work; the predictors are **heuristic group-additivity** models, suitable for educational use and qualitative functional-group identification, with the limitations stated in the *Validation Report*.

1. The architecture is **local-first**: no user personal data is transmitted to author-controlled servers (alignment with GDPR principles).

This is an **author's declaration**, not a certification issued by a third party.

6. Attached documentation

1. **Software Architecture Document (SAD)** — architecture, components, technologies.
2. **Verification & Validation & Benchmark Report** — test method, benchmarks, limitations.
3. **Security, Privacy & Compliance** — local-first, GDPR, OWASP, GAMP 5 (alignment).
4. **Open-Source Licensing & IP Matrix** — dependencies, licenses, intellectual property.

Software license: proprietary — "All rights reserved" (see **LICENSE**).

Software Architecture Document (SAD) — BioSpecInfo

Field	Value
Project	BioSpecInfo
Author	Samuele Pio Provenzano
Thesis advisor	Prof. Savino Longo — University of Bari Aldo Moro
Type	Progressive Web App (PWA), client-side, local-first, offline-first
Platform	Modern browsers (Chromium, Firefox, Safari/WebKit) · Desktop and Mobile
Repository	<code>samupropio1-ship-it/BioSpecInfo-v11</code>

1. System overview

BioSpecInfo is a **monolithic yet modular** web platform that runs all of its application logic inside the client browser, with no application server. It is distributed as an installable **Progressive Web App** (`display: standalone`) and works **offline** thanks to a Service Worker that precaches the heavy assets.

Architectural pattern: Single-Page/Single-File Application + satellite sub-apps in iframes, with a WebAssembly-based cheminformatics *core* decoupled from the presentation logic.

1.1 Guiding principles

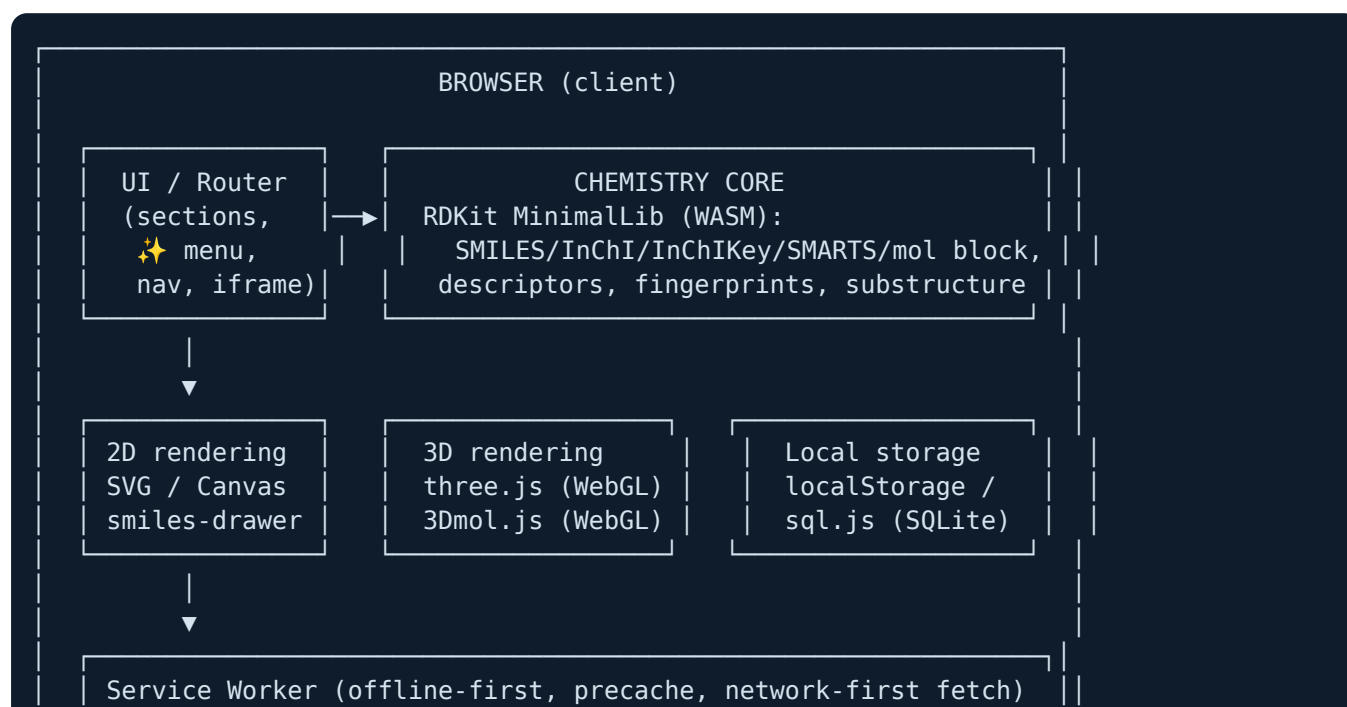
- **Local-First:** data and computations live on the device; no mandatory transmission to third parties.
- **Zero-backend:** no server database, no dynamic hosting cost (deployable on static hosting, e.g. GitHub Pages).
- **Progressive enhancement:** online features (PubChem search, real 3D models, astronomical imagery) degrade gracefully when offline.

2. File structure

File	Role
<code>index.html</code>	Main application (chemistry core, Spectra Centre, Data Science, ✨ menu). ~4.5 MB, over 43,000 lines.
<code>chimorga.html</code>	Organic Chemistry course (25 modules).
<code>astro.html</code>	3D Astrochemistry (three.js).

File	Role
pro.html, rdkit_lab.html, accademia.html, simulazioni.html, guidaret.html, file_manager.html, Biochimica_Guida_Definitiva.html, sr_completo.html, sr_essenziale.html, download.html, changelog_tesi.html	Thematic sub-apps loaded in iframes with ?v=timestamp cache-busting.
manifest.json	PWA configuration (icons, theme, standalone).
sw.js	Service Worker: <i>network-first</i> strategy with cache fallback; precache of heavy libraries/assets; cache version bumped on every release.
RDKit_minimal.js / RDKit_minimal.wasm	Cheminformatics engine (RDKit MinimalLib).
3Dmol-min.js	3D viewer for molecules/macromolecules.
three.min.js, three_bloom.js, gltf_loader.js	WebGL 3D rendering (anatomy, astrochemistry, mechanisms).
smiles-drawer.min.js	2D depiction of SMILES.
lib/sql-wasm.js / lib/sql-wasm.wasm	SQLite in WebAssembly (SQL playground).
models/, textures/	3D assets (organs, écorché, planets) and textures.

3. Component architecture



| (online features only, optional)

Public APIs: PubChem PUG-REST · NASA/ESA · links to spectral databases

3.1 Module Router / UI

Navigation between sections via CSS classes (`.section.on`) and a ✨ menu (FAB) that opens the special tools (Spectra Centre, Retrosynthesis, Quantum Lab, Data Science, etc.). Sub-apps are isolated in versioned `iframes`.

3.2 WebAssembly Engine (RDKit MinimalLib)

Loaded on demand (`window.bsiLoadRDKit`) with `locateFile` pointing to the local `.wasm` files and a CDN as a last-resort fallback. It exposes:

- **Identifiers:** canonical/isomeric SMILES, CXSMILES, InChI, InChIKey, SMARTS,

MDL V2000 mol block.

- **Descriptors:** molecular weight, monoisotopic exact mass, LogP

(Wildman–Crippen), TPSA, HBD/HBA, rotatable bonds, aromatic/aliphatic rings, Fsp³, molar refractivity, stereocentres.

- **Fingerprints & similarity:** Morgan ECFP4 (1024 bit) and the Tanimoto index.
- **Substructure matching** via SMARTS (`get_qmol` + `get_substruct_matches`).
- **2D depiction** (`get_svg` , `get_svg_with_highlights`) and 2D coordinate

generation (mol block) re-importable into the editor.

3.3 Rendering

- **2D:** proprietary SVG (reaction mechanisms with Bézier arrows), Canvas 2D

(molecule editor, CIP stereochemistry viewer), smiles-drawer.

- **3D:** `three.js` (WebGL) for anatomy (Drug Atlas), astrochemistry and 3D

mechanisms; `3Dmol.js` for molecular models from PubChem SDF.

3.4 Storage Layer

- `localStorage` (with a **safety guard** and in-memory fallback for incognito /

blocked-storage scenarios).

- `sql.js` (SQLite compiled to WASM) for the Data Science SQL playground.

4. Implemented features (current state)

4.1 Spectra Centre (molecular editor & analysis)

Draw from scratch (full periodic table, ring/group templates, drag-to-bond with snapping) or load via SMILES / library (63 molecules) → full analysis:

- **All chemical notations** (formula, canonical SMILES, CXSMILES, SMARTS, InChI, InChIKey, mol block, common/IUPAC name, CAS) with one-click copy.
- **Predicted spectra** as sub-panels: IR, ^1H -NMR, ^{13}C -NMR, MS with ****isotope pattern (IUPAC abundances) and adduct masses**** ($[\text{M}+\text{H}]^+$, $[\text{M}+\text{Na}]^+$...), UV-Vis, Raman — each with a bibliographic source.
- **Functional groups** (~30, via SMARTS) and **properties** (H donors/acceptors, aromatic, rotatable, stereocentres) **highlightable on the 2D and 3D structure**.
- **Drug-likeness / ADMET**: Lipinski, Veber, Egan, Ghose, Muegge rules; estimated ESOL solubility; QED (when exposed by the engine); **Brenk/PAINS structural alerts**; physicochemical radar profile.
- **Elemental analysis** (weight %) and **Tanimoto similarity** against the library.
- **3D models** (ball & stick, spacefill, wireframe) from PubChem; SVG/PNG/MOL export.

4.2 Synthesis & Retrosynthesis

46 disconnection strategies with an **animated mechanism player** in **2D and 3D**: curved *electron-pushing* arrows, reaction intermediates, and a retrosynthesis mode showing **synthon formation**.

4.3 Organic Chemistry ([chimorga.html](#))

25 modules: fundamentals, stereochemistry, alkenes/alkynes, aromatics & EAS, SN/E, carbonyls, amines, carbohydrates, amino acids, lipids, nucleotides, animated syntheses, named reactions, heterocycles, pericyclic & Woodward–Hoffmann, organometallics & cross-coupling, radicals, spectroscopy, retrosynthesis & protecting groups.

4.4 Data Science

3D Drug Atlas (13 body regions, 63 diseases of which 23 tumours, 3D organ models, PubMed links), interactive ML labs, SQL playground (sql.js), course roadmap.

4.5 Other sections

3D astrochemistry, Quantum Lab, Academy (gamified quizzes with SM-2 spaced repetition), interactive simulations, guides, databases of elements/drugs/ biomolecules.

5. Architectural roadmap (NOT yet implemented)

The following items belong to the *enterprise/HPC* vision and require external binaries, models or build pipelines that cannot be hosted inside a single-file PWA:

- HPC parallelism via Web Workers + [SharedArrayBuffer / Atomics](#) .
- WebGPU acceleration for dense/sparse algebra and GNN inference.
- ML predictors in ONNX Runtime WebGPU (GNN for NMR such as ShiftML2/DeepNMR; hERG, BBB, CYP450, LogS).
- In-browser molecular docking (AutoDock Vina/Webina, PDBQT format).

- Semi-empirical quantum chemistry (xTB/GFN2-xTB in WASM).
- Rule-based computational retrosynthesis engine (SMIRKS).
- SQLite-WASM database at the 1,000,000-structure scale and SDF batch processing

via Web Workers.

- Immersive WebXR (VR/AR) visualisation.
- Native packaging (Android TWA, iOS, Electron/Tauri) and CI/CD pipelines.

6. Requirements and constraints

- **Browser:** WebAssembly and WebGL support; JavaScript enabled.
- **Network:** not required for the core; recommended for PubChem search and real

3D models.

- **Performance:** target 30–60 fps for interactive rendering; heavy computations are delegated to RDKit WASM and kept responsive.

Verification & Validation (V&V) & Benchmark Report

— BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Purpose	Document the verification method, validation data for the spectral predictors, cross-device performance and declared model limitations.

1. Scope and V&V strategy

- **Verification:** the software is built correctly (syntax, no runtime errors, expected component behaviour).
- **Validation:** the scientific results (spectral predictors, descriptors) are compared against reference experimental databases.

2. Software verification (QA)

Technique	Description	Status
Syntax checking	<code>node --check</code> on the critical JavaScript blocks on every change.	✓ In use
End-to-end (E2E) testing	Automation with Playwright on headless Chromium: open sections, draw/load molecules, verify panels (notations, spectra, functional groups, ADMET, similarity), 2D/3D animations. Counting of console/page JavaScript errors (acceptance criterion: 0 non-network errors).	✓ In use
HTTP-server verification	RDKit WASM requires an HTTP context (CORS): tests run on a local server to reproduce the real environment.	✓ In use
Manual cross-device testing	Testing on real devices (see §4).	✓ In use
Automated unit-test suite (Vitest/Jest)	—	 Roadmap
Coverage ≥ 90%	—	 Roadmap

Every release is verified E2E with Playwright under a "zero JS errors" criterion before merging to the main branch.

3. Scientific validation of the spectral predictors

BioSpecInfo's spectral predictors use **heuristic group-additivity / structural rule** models (literature reference tables), chosen to guarantee **instant in-browser response** on any device.

3.1 ¹H-NMR benchmark vs SDBS

Metric	Reported value
Reference database	SDBS (Spectral Database for Organic Compounds, AIST)
Mean Absolute Error (MAE)	0.31 ppm
Signals within ± 0.5 ppm	89%

3.2 IR benchmark (C=O band) vs NIST

Metric	Reported value
Reference database	NIST Chemistry WebBook
Mean Absolute Error (MAE)	11 cm⁻¹
Bands within ± 30 cm ⁻¹	90%

3.3 Physicochemical properties

Descriptors (molecular weight, formula, LogP, TPSA, HBD/HBA, rings, stereocentres) are computed with **RDKit MinimalLib** and comparable with **PubChem** and **DrugBank**. Reproducible verification example: *cholesterol* $\rightarrow$ $C_{27}H_{46}O$, 4 rings, 8 stereocentres.

Benchmark data source: measurements carried out by the author as part of the undergraduate thesis. The values are to be understood as validation for **educational and qualitative functional-group identification** purposes, not as a substitute for reference quantum-mechanical calculations.

4. Cross-device performance

Smoothness tests of the interactive renderers (3D viewers, animations, editor) across a range of devices:

Device	Result
iPhone / iPad	30–60 fps
Windows / macOS laptop	60 fps
Raspberry Pi 4	30–60 fps

Size of the main `index.html` file: $\approx$ **4.5 MB** (the full package, including WASM modules and 3D assets, is larger but remains entirely static and cacheable offline).

5. Declared model limitations

In line with the principle of scientific transparency:

1. **Additivity model.** The NMR/IR predictors sum group contributions and do **not** fully model extended chemical-environment effects (long conjugation, anisotropic effects, intramolecular hydrogen bonds). Larger deviations are expected e.g. for **strongly conjugated carbonyls** or edge cases such as **formic acid**.

1. **Qualitative/educational spectra.** The simulated spectra serve to recognise functional groups and understand structure–spectrum relationships, not to replace certified experimental data.

1. **Network dependency for some functions.** IUPAC/common name, CAS number, GHS pictograms and real 3D models come from PubChem: without network the app shows a fallback and remains fully functional on local data.

1. **Two-way accuracy.** For rigorous use, direct links to the official databases (NIST, SDBS, MassBank, nmrshiftdb2) are integrated into the interface.

6. Release acceptance criteria

-  `node --check` passes without errors on the modified modules.
-  Playwright E2E: 0 non-network JS errors.
-  No visible regression in the touched sections.
-  Service Worker cache version bumped.

Security, Privacy & Compliance — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Model	Client-side / Local-First PWA

1. Privacy & data protection (GDPR)

1.1 Local-First by design

- **No account, no login, no advertising tracking.**
- All user data (drawn molecules, preferences, quiz history, study progress) is

stored **exclusively locally** on the device (`localStorage` / IndexedDB / `sql.js`), with no transmission to author-controlled servers.

- **There is no application backend** collecting or processing personal data.

1.2 Data handling and GDPR alignment

- **Data minimisation:** the app requires no personal data to function.
- **No transfer to the author's third parties:** data never leaves the device for

infrastructure controlled by the author.

- **Optional, transparent external calls:** only upon a user action (searching a molecule, opening a 3D model) does the app query public third-party APIs (PubChem, NASA/ESA, spectral databases). Such requests contain **only the chemical identifier** (e.g. SMILES/name) and **no personal data**. Use of these services is subject to their respective privacy policies.

- **Local right to erasure:** the user can delete data by clearing the site storage in the browser.

Note. This alignment describes the local-first design properties that favour GDPR compliance. It does not constitute a legal certification; for regulated contexts a dedicated DPIA assessment is recommended.

2. Application security (OWASP)

2.1 Attack surface

Being a **static application with no backend**, entire classes of server-side vulnerabilities are absent by construction (SQL injection on a server DB, server-side RCE, server authentication/session, IDOR on proprietary APIs).

2.2 Input sanitisation

- Parsing of **SMILES / SMARTS** strings and molecular files is delegated to

RDKit MinimalLib (WASM), which validates the structures and rejects malformed input by returning invalid molecules rather than executing them.

- Strings coming from user input or external APIs, when inserted into the DOM,

are treated as untrusted. Their textual value (names, notations) is shown as text content/ `textContent` or with escaping of the `& < >` characters.

2.3 Runtime robustness

- **localStorage guard**: every storage access is protected by `try/catch` with

an in-memory fallback, so that incognito mode or blocked storage cannot halt execution (a historical regression that has been fixed).

- **Controlled degradation**: panels never silently stay empty; errors show a

message and, where applicable, a "Retry" button.

- **Sub-app isolation** in `iframes`.

2.4 Content Security Policy (deployment recommendation)

Since the app can be published on static hosting, it is recommended to serve it with security headers at the hosting/CDN level:

- Restrictive `Content-Security-Policy` (self script/style + WASM),

`frame-ancestors`, `X-Content-Type-Options: nosniff`, `Referrer-Policy: no-referrer`.

- `Cross-Origin-Opener-Policy` / `Cross-Origin-Embedder-Policy` if

`SharedArrayBuffer` and parallel Web Workers are enabled in the future (HPC roadmap).

3. Integrity and reproducibility (GAMP 5 alignment)

For use in pharmaceutical/regulated contexts, the architecture is **compatible** with GAMP 5 principles:

- **Determinism**: the cheminformatics computations (RDKit) are deterministic and

reproducible for the same input.

- **Version traceability**: code versioning on Git, versioned Service Worker

cache, changelog.

- **Method transparency**: the heuristic predictors are documented with their

sources and limitations (see the *Validation Report*).

Again: this is **alignment with the principles**, not a formal qualification (IQ/OQ/PQ), which would require a dedicated validation process at the using organisation.

4. Summary

Aspect	Status
Personal data transmitted to author's servers	None
Backend / server database	Absent
Molecular input validation	RDKit WASM
DOM output escaping	Yes (untrusted content)
Storage guard / error degradation	Yes
CSP / security headers	Recommended at hosting level
Offline operation	Yes (Service Worker)

Open-Source Licensing & IP Compliance Matrix — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author / owner of proprietary code	Samuele Pio Provenzano

1. Third-party dependency matrix

All bundled third-party libraries are **open-source under permissive licenses** (MIT / BSD-3-Clause). **No viral copyleft dependency (GPL / AGPL / LGPL)** is used, so there are no obligations to redistribute the proprietary code.

Component	Repo file	Use	License
RDKit (MinimalLib / rdkit-js)	RDKit_minimal.js , RDKit_minimal.wasm	Cheminformatics engine (SMILES/InChI/InChIKey/SMARTS, descriptors, fingerprints, 2D depiction)	BSD-3-Clause
3Dmol.js	3Dmol-min.js	3D viewer for molecules/macromolecules (WebGL)	BSD-3-Clause
three.js	three.min.js , three_bloom.js , glTF_loader.js	WebGL 3D rendering (anatomy, astrochemistry, mechanisms)	MIT
SmilesDrawer	smiles-drawer.min.js	2D depiction of SMILES strings	MIT
sql.js (SQLite compiled to WASM)	lib/sql-wasm.js , lib/sql-wasm.wasm	Client-side SQL playground	MIT

The exact versions are those of the files included in the repository. Each permissive license requires preserving the original copyright notice, which must be kept in the library files or in a `THIRD_PARTY_NOTICES` file.

2. External services/APIs (not redistributed)

These services are queried over the network at runtime (their data is not part of the package); their use is subject to the respective terms.

Service	Use
PubChem PUG-REST (NIH/NCBI)	IUPAC/common name, CAS, formula, 3D SDF models, GHS/CLP pictograms

Service	Use
NIST WebBook · SDBS · MassBank · nmrshiftdb2 · SpectraBase	Links to reference experimental spectra
NASA / ESA	Astronomical images and data (Astrochemistry section)

3. Intellectual property of the proprietary code

• BioSpecInfo's application code (UI logic, the proprietary SVG mechanism engine, the CIP stereochemistry viewer in Canvas, the heuristic spectral predictors, the educational content, the internal databases) is the **author's original work**, except for the third-party libraries listed above.

• **No copyleft-licensed code** is embedded in the proprietary source: the author retains full licensing freedom over their own code.

3.1 Project license

The repository includes a **proprietary — "All rights reserved"** [LICENSE](#) file: the code is viewable and assessable (personal study and professional/academic evaluation), but **copying, modification, redistribution and commercial use require the Author's written permission**. This choice protects the project from unauthorised reuse while keeping it presentable to recruiters and companies. Third-party components remain governed by their permissive licenses.

4. Compliance summary

Item	Result
Viral copyleft dependencies (GPL/AGPL)	None
Dependency licenses	Permissive (MIT / BSD-3-Clause)
Obligation to keep copyright notices	Yes (in library files / NOTICES)
Licensing freedom over proprietary code	Full
Project LICENSE file	 Present — proprietary "All rights reserved"